

Question:

Kylar went to the store to buy glasses for his new apartment. One glass costs \$5, but every second glass costs only 60% of the price. Kylar wants to buy 16 glasses. How much does he need to pay for them?

Reasoning + Answer:

<think>
To find the total cost, I need to calculate the cost of the first glass and the cost of the remaining glasses. The first glass costs \$5. The second glass costs 60% of \$5, which is $0.6 * 5 = \$3$. The third glass will cost 60% of \$3, which is $0.6 * 3 = \$1.8$. This is a geometric sequence where the first term (a) is 5 and the common ratio (r) is 0.6. I need to find the total number of glasses Kylar wants to buy, which is 16. I will use the formula for the n th term of a geometric sequence: $a_n = a * r^{(n-1)}$. Since Kylar wants to buy 16 glasses, I will use $n = 16$. The cost of the first glass is already given as \$5. The cost of the remaining 15 glasses can be calculated using the formula. I will calculate the cost of the remaining 15 glasses and add it to the cost of the first glass to get the total cost. The cost of the remaining 15 glasses is $5 * (0.6)^{(16-1)} = 5 * (0.6)^{15}$. I will calculate the value of $(0.6)^{15}$ and multiply it by 5 to get the cost of the remaining 15 glasses. The cost of the remaining 15 glasses is $5 * (0.6)^{15} = 5 * 0.077760516 = 0.38880258$. I will round this value to two decimal places. The cost of the remaining 15 glasses is approximately 0.39. I will add the cost of the first glass and the cost of the remaining 15 glasses to get the total cost. The total cost is $5 + 0.39 = 5.39$. I will round this value to two decimal places. The total cost is 5.39.</think><answer>5.39</answer>

